BSH/W-23/CS/AI/DS1T001/02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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Education to Eternity

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

B.Tech / ~~M.Tech~~ / ~~MBA~~ / ~~MCA~~ ~~Summer~~ / Winter - 2023

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch:CSE/AI/CSE(DS) Sem:-I

Subject: Numerical Method & Advance Calculus

Subject Code:CS1T001/AI1T001/DS1T001

Marks: 100 **Date of Exam:** 27/12/2023 **Time: 3 Hrs.** BT23000LCS

Instructions:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

COs: Students should be able to:

1. Describe concept of linear algebra, probability, logic, numerical method, distribution theory.
2. Illustrate the concept of linear algebra, probability, logic, numerical method, distribution theory by using examples.
3. Apply the knowledge of linear algebra, probability, logic, numerical method, distribution theory to solve the engineering problems.
4. Analyze the problems and results of linear algebra, probability, logic, numerical method, distribution theory to solve the real world
5. Evaluate the problems by linear algebra, probability, logic, numerical method, distribution theory.
6. Create the methods or model by linear algebra, probability, logic, numerical method, distribution theory.

PARTA

Q. No.	Solve the two words or two sentences (All Questions are compulsory).	Marks	Bloom's Level and CO
Q.1	What is the statement of Bay's Theorem.	2	L1/CO1
Q.2	Explain mutually exclusive events.	2	L2/CO2
Q.3	Tell the formula of Trapezoidal rule.	2	L1/CO1
Q.4	Tell the formula of Simpson's 1/3 rule.	2	L1/CO1
Q.5	Define Curl F.	2	L1/CO1
Q.6	What is mean by coplanar vector.	2	L1/CO1
Q.7	Define Conjunction.	2	L1/CO1
Q.8	Illustrate the frequency Polygon	2	L2/CO2
Q.9	What is man by frequency	2	L1/CO1
Q.10	Tell the formula of distribution function of continuous random variable	2	L1/CO1
Q.11	Explain Union of sets.	2	L2/CO2
Q.12	What is mean by Logical Equivalence.	2	L1/CO1

PART B

Q.No.	Solve the following Descriptive Type Questions	Marks	Bloom's

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(All Questions are compulsory).

		Marks	Level and CO
Q.13	A random variable X has the probability function $f(x) = ce^{-3x}, x \geq 0$ $= 0,\ x < 0$. Solve to find the constant c .	4	L3/CO3
Q.14	From the integration $\int_0^2 (e^x . sinx)\ dx$. Illustrate the corresponding value of x ,taking 4 division.	4	L2/CO2
Q.15	If $\emptyset = xy + y^2x + z^2$. Construct grad $\emptyset$	4	L3/CO3
Q.16	Explain the negation of the statement " If he work, he will earn money".	4	L2/CO2
Q.17	Summarize the characteristics of the Histogram.	4	L2/CO2
Q.18	Construct the given statement is logically equivalence $\sim(p \leftrightarrow q) \equiv q \leftrightarrow \sim p$.	4	L3/CO3

PART C

Q. No.	Solve the following Descriptive Type Questions (Solve any 04 out of 06).	Marks	Bloom's Level and CO
Q.19	A factory manufacturing televisions has four units A,B,C,D.The unit manufacture 15%, 20%,30% and 35% of total output respectively It was found that out of their outputs 1%, 2% ,2% ,and 3% are defective.A television is chosen at random from the total output and was found to be defective . Analyze the probability that it was manufactured by unit C.	7	L4/CO4
Q.20	Solve $\int_0^4 e^x\ dx$ using Simpson's 1/3 rule using four strips.	7	L3/CO3
Q.21	Simplify that A is irrotational if $A = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - y)k$ And hence find $\emptyset$ such that $A = \nabla\emptyset$	7	L4/CO4
Q.22	Construct that $A - (B \cup C) = (A - B) \cap (A - C)$	7	L3/CO3
Q.23	Explain Line graph with example.	7	L2/CO2
Q.24	A random variable X has the following probability function	7	L3/CO3

X	0	1	2	3	4	5	6	7	8
f(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

Solve to find (i) value of a (ii) P (X< 3)

PART D

Q. No.	Solve the following Descriptive Type Questions (Solve any 02 out of 03).	Marks	Bloom's Level and CO
Q.25	A random variable X has density function $f(x) = kx^2, 1 \leq x \leq 2$ $= kx, 2 \leq x \leq 3$ $= 0, otherwise$. Examine the value of k and distribution function	12	L4/CO4
Q.26	Gas is expanded according to low $PV^{1.3} = C$ from the pressure of $10\ N/m^2$. Assuming initial volume of gas $1\ m^3$ and final volume $7\ m^3$. Construct work done using Simpson's 1/3 rule. Divide volume in 6 equal strips.	12	L3/CO3
Q.27	A vector field F is given by $F = (ysinz - sinx)i + (xsinz + 2yz)j + (xycosz + y^2)k$. Simplify that it is irrotational and hence find its scalar potential.	12	L4/CO4

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